



CENTER OF EXCELLENCE IN MANUFACTURING ANALYTICS



Walmart 



Walmart Center for Tech Excellence

ABOUT THE CoE



Prof. Raghunathan Rengaswamy
Head-Walmart Center for Tech Excellence



MoU signing between IITM and Walmart Global Tech

The Walmart Center for Tech Excellence is an initiative aimed at fostering innovation, research, and technological advancements in collaboration with industry and academic partners. It focuses on areas such as AI, automation, supply chain optimization, and digital transformation to drive efficiency and innovation. The CoE focuses on digital empowerment of the MSME operating in manufacturing, and aligned sectors. The Center delivers cutting-edge AI solutions to boost the enterprises' operational efficiencies and profitability. The Center also facilitates skill development initiatives to accelerate the scalable deployment of these solutions. The collaborating team comprises of industry experts, core faculty members from various institutes, Ph.D. students, post-doctoral fellows and project associates. The CoE provides valuable insights, resources, and potential partnerships that aligns with the strategic goals.

Connect with us

Landline: 044 - 2257 9465

email: connect.wcte@wcte.iitm.ac.in

GOALS

- Development of a domain-agnostic platform that will enable MSME segments to develop engineering analytics solutions.
- Development of use case studies repository that will accelerate AI adoption in manufacturing industries (MSME).
- Development and delivery of skill-development programs on AI and IoT and its use in manufacturing for MSME segments.

OUR VISION

Develop affordable, easily deployable, and maintainable AI/ML and digitalization solutions for manufacturing and retail businesses through cutting-edge scalable open-source platforms and IoT.

OUR MISSION

Build solutions and world-class educational material for micro, small, and medium enterprises (MSME) to adopt AI/ML technologies and digitization toward improved operational efficiency and profit margins.

OUR APPROACH

- **Industry Collaboration:** We partner with industry experts to develop solutions tailored to the unique challenges faced by MSMEs in the manufacturing sector .
- **Academic Partnership:** Collaborating with researchers from IIT Madras, we ensure our solutions are grounded in the latest research and innovations.
- **Holistic Solutions:** We address common challenges such as fragmented data sources and skill gaps.

PRODUCTS DEVELOPED AT THE CoE

PLATFORMS DEVELOPED

DATAPY

- Managerial dashboard
- Visualizations
- Basic data analytics

FABRIC (Mini ERP)

- Shop floor order tracking
- Inventory management
- Planning

MANGO

- Computer vision solutions
- Flow-based drag-drop
- Customizable templates for advanced AI/ML analytics

TOOLS DEVELOPED

PREDICTIVE MAINTENANCE (PdM)

The PdM system monitors key parameters to predict rotary asset failures like bearing faults, preventing unplanned downtime and costly maintenance.

NON-CONFORMANCE REDRESSAL (NCR) SYSTEM

The NCR system digitizes legacy records and leverages AI-driven, agent-assisted tools to help MSMEs.

OPTICAL CHARACTER RECOGNITION (OCR)

Optical Character Recognition (OCR) is a technology that converts various types of documents such as scanned paper documents, PDF files, or image captured by a digital camera into editable and searchable data.

KAATRU

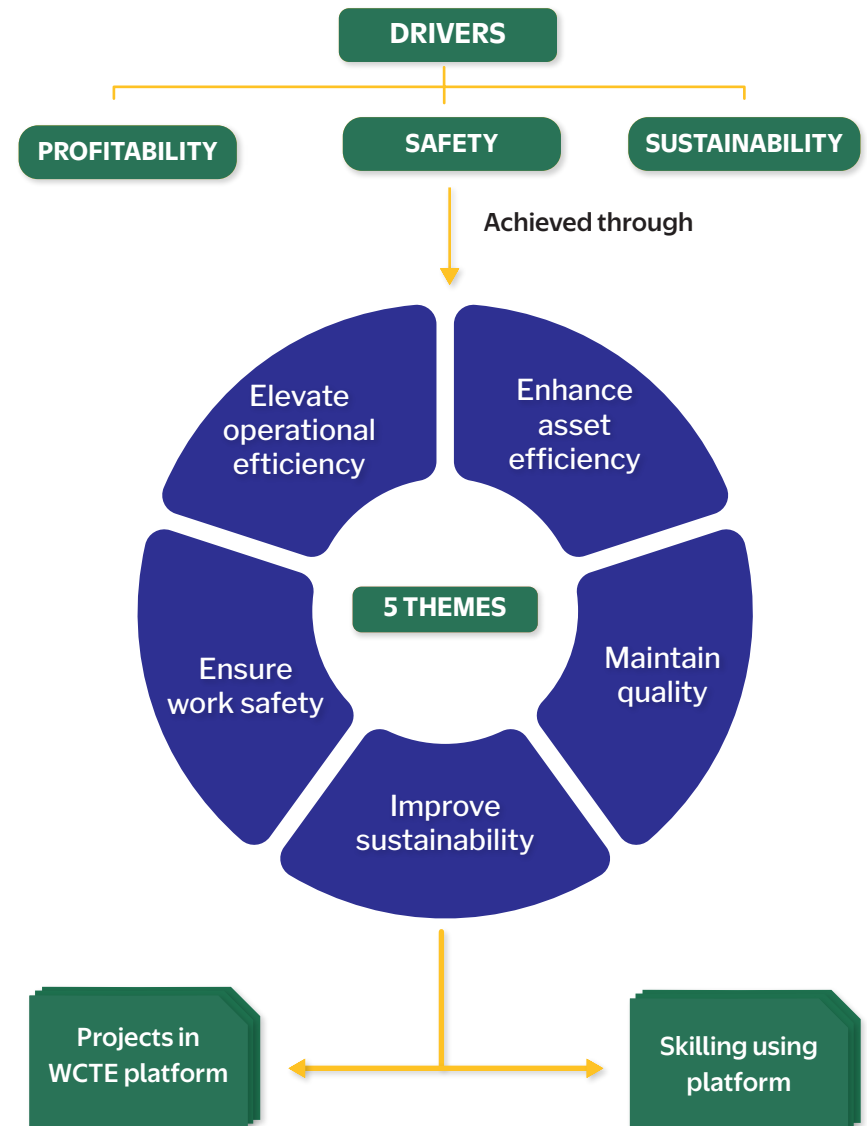
Kaatri is a hyperlocal air quality tool dedicated to providing real-time environmental insights that empower individuals to make informed health decisions.

For Project Collaborations

Landline: 044 - 2257 9463

email: project.support@wcte.iitm.ac.in

THEMES DRIVEN BY THE GOALS



PLATFORMS



DATAPY

DATAPY is a cloud-enabled data visualization and analytics platform that transforms raw data from diverse sources like Google Sheets into interactive dashboards. It empowers businesses to track performance, uncover patterns, and make data-driven decisions with ease.

Key Features

1. Interactive visualizations-converts raw data into intuitive dashboards
2. Drill-down approach-navigate from high-level summaries to detailed insights
3. Custom graphs & plots-compare key performance metrics visually
4. Production bottleneck detection-pinpoint inefficiencies quickly
5. User-friendly interface - easy access to resource groups and history

Benefits

1. Moves organizations from reactive to proactive decision-making
2. Optimizes inventory and production schedules
3. Strengthens customer relationships with better service insights
4. Provides a complete operational picture in just a few clicks
5. Enhances efficiency and transparency across business processes



FABRIC

Fabric is a cloud based ERP solution designed for MSMEs to streamline order management, workflows, and inventory. It automates routine processes, reduces manual inefficiencies, and ensures real-time visibility across operations.

Key Features

1. PO-centric approach for batch planning & tracking
2. Automated batch processing to reduce errors
3. Role-based access for secure operations
4. Inventory & RFQ management with digital records
5. Quotation generation & automated communication

Benefits

1. Real- time tracking across various processes
2. Faster decision making
3. Improves operations productivity
4. End- to End tracking of orders
5. Cost efficiency minimizes waste and ensures timely dispatch



MANGO

MANGO (Manufacturing Analytics Navigating Growth and Opportunities) is a flow-based data science platform that simplifies the creation and execution of AI/ML workflows. It enables users to design workflows through a drag-and-drop studio with pre-built blocks for data preparation, analysis, and visualization.

Key Features

1. Flow-based design studio for visual ML workflows
2. Drag-and-drop interface for workflow customization
3. Pre-built templates & reusable components
4. Cloud execution using docker containers
5. Blocks for data sourcing, cleaning, transformation, analysis & visualization

Benefits

1. Reduces technical complexity by making MANGO user friendly
2. Boosts productivity with intuitive workflow design
3. Accelerates AI/ML adoption through ready-to-use templates
4. Improves scalability, flexibility & deployment speed
5. Enhances decision-making with simplified data-driven workflows

TOOLS



NON-CONFORMANCE REDRESSAL (NCR) SYSTEM digitizes legacy paper-based records and empowers with AI-driven, agent-assisted tools for analyzing non-conformances.

Key Features

1. Document digitization of NC records
2. Smart autocompletion of forms
3. Knowledge Graph+ LLM interface for guided analysis
4. Detailed root cause analysis & insights
5. Analytics on severity, frequency & recurrence

Benefits

1. Low-cost auto digitization
2. Reduces report generation
3. Proactively detects recurring issues through pattern analysis
4. Reduces manual work and improves accuracy
5. Strengthens adherence to quality standards



PREDICTIVE MAINTENANCE (PdM) is designed to streamline the monitoring and analysis of rotary assets with bearings using sensors, helping reduce unplanned downtime in manufacturing plants.

Key Features

1. Real time monitoring
2. Summary of the health status of the assets with condition based color coding
3. Visualizations for fault severity, temperature & RMS
4. Gauge charts for an overall view of the threshold limits
5. Weekly comparison charts for severity, temperature and RMS

Benefits

1. Early fault detection with alarm setup
2. Reduced unplanned downtime
3. Planned maintenance of assets and inventory
4. Optimum utilization of resources
5. Improves RUL of assets



OPTICAL CHARACTER RECOGNITION (OCR) is a tool that converts various types of documents such as scanned paper documents, PDF files, or image captured by a digital camera into editable and searchable data.

Key Features

1. Converts physical paper records into high-quality and fully searchable digital data
2. Identifies and captures specific data fields from invoices and purchase orders
3. Seamless integration enable auto- export to standard formats like Excel or CSV
4. OCR ensures documents have a secure, long- lasting digital copy
5. Digital records can be accessed from multiple devices, anytime and anywhere

Benefits

1. The tool maximizes efficiency by processing forms and documents faster
2. Actionable analytics for better decision making
3. Cloud capabilities enhance accessibility of documents securely from anywhere
4. Reduces operational costs and minimizes human error
5. Improves workflow visibility



KAATRU is a hyperlocal air quality tool that transforms accurate pollution data into actionable awareness, and sustainable communities through increased environmental transparency.

Key Features

1. Development of stationary air quality monitoring devices
2. Generate reports for the industries on their current environment quality
3. Provide environmental mitigation strategies and policies
4. Interactive dashboards to visualize real-time environmental and historical data
5. Foster to build more healthier communities

Benefits

1. Access to real time air quality index
2. Historic records of air quality trends
3. Reports with seasonal variation and severity analysis
4. Data driven actionable insights
5. Industry and environmental pollution transparency is recorded

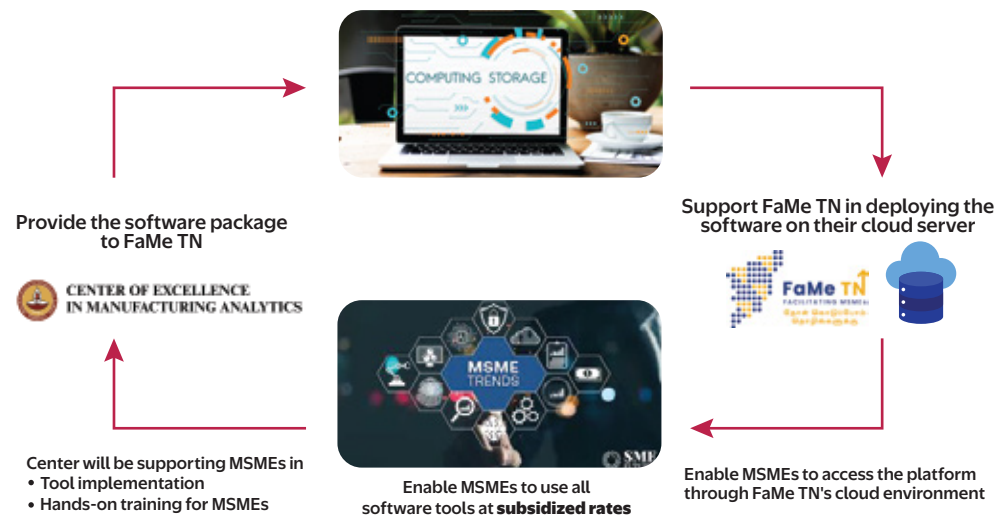
WORKFORCE DEVELOPMENT PARTNERSHIP WITH FaMe TN



MoU signing between IITM and FaMe TN

The overarching objective of the collaboration with FaMe TN (Facilitating MSMEs of Tamil Nadu) is to strengthen the digital and technological capabilities of MSMEs across Tamil Nadu. Through this partnership, targeted training programs, workshops, and awareness sessions are conducted to promote Industry 4.0 adoption, smart manufacturing, and data-driven decision-making.

The initiative focuses on capacity building, technology enablement, and skill development, supporting MSMEs to improve productivity, operational efficiency, and competitiveness in an increasingly digital industrial ecosystem.



For Skill & Development
Landline: 044 - 2257 9468
email: training.wcte@wcte.iitm.ac.in

SKILLING AND TRAINING

The CoE organizes a series of training sessions designed to create awareness and build practical skills in Industry 4.0, IoT, AI/ML, and digital manufacturing. These sessions aim to bridge the gap between industry needs and emerging technologies.

MSMEs

The training focuses on enabling MSMEs to understand and adopt smart manufacturing practices. Topics included predictive maintenance, process optimization, data analytics, and digital transformation strategies. Participants gain insights into how these technologies can improve productivity, reduce downtime, and enhance competitiveness.

Faculty and Students

These sessions emphasize hands-on learning through live demonstrations, real-world case studies, and interactive discussions. The program provides an overview of IoT ecosystems, the integration of AI in manufacturing, and the fundamentals of data-driven decision-making. This helps students prepare for careers in advanced manufacturing and industrial analytics.



STAKEHOLDERS TESTIMONIALS

Mr. Ranjith R

General Manager, Amma Alloys, Coimbatore



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We are pleased to collaborate with such a high-caliber professional team to implement Industry 4.0 solutions for predictive maintenance. In today's competitive landscape, this initiative aims to reduce breakdowns and prevent production losses. It serves as a significant opportunity for MSMEs to adopt advanced technologies and achieve enhanced productivity without hindrance. We are excited to be part of this transformative process.

Mr. Gopalakrishna Hegde,

Managing partner, Shreyas Machine Tools, Bengaluru

SMT

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Working with this professional team has been a game-changer for our business. Their deep understanding of customer pain points, combined with their expertise in developing affordable, user-friendly solutions, has made a significant impact. They focus on up-skilling, ensuring that the solutions they provide are not only cutting-edge but also practical for MSMEs. By leveraging the latest technologies like AI and data science, they've delivered products that truly make a difference for both my business and others in the MSME sector.

Mr. Arun Ramu,

Founder & CEO, MachineMaze, Bengaluru



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Extremely relevant to accelerate India's manufacturing needs and improve utilization of our MSME ecosystem

Mr. Parthasarathy. T. R,

Founder & CEO, Hustle Minds Solutions Pvt Ltd, Chennai



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Great to see AI and manufacturing analytics being built with real business usability in mind. Strong example of industry-academia collaboration creating practical impact for MSMEs.

Mr. Purushotham S Joshi,

Admin & Planning, Accutech Laser Pvt. Ltd, Bengaluru



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Working with this professional team has been an absolute pleasure. They truly understand our needs and consistently deliver products that not only meet, but exceed expectations. Their commitment to quality and stake holder satisfaction ensures that the end user always benefits.

Ms. Deepika Prabhakaran,

Intern - Walmart Center for Tech Excellence, Chennai

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I attended the MANGO Workshop conducted by the Walmart Center for Tech Excellence at IIT Madras, which led to my selection as an AI Research Intern. Since then, I've been actively learning, experimenting, and building innovative solutions in a collaborative, high-growth environment.

Ms. P. S. Amudha,

Sole proprietor, G.R.Mart, Chennai



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The workshop gave me new insights to minimise human work & avoid human errors. Thanks for a very interactive session and for the hands-on practice with the tools. More user-friendly tools to minimise our paperwork and human resources.

Ms. Radhika Subramanian,

Founder, Ermunai Organic Farm & Foods, Chennai



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The training programme delivered practical, industry-focused insights into process improvement and digital transformation. The learnings have been directly applicable to our work, helping us improve efficiency, productivity, and accuracy across operations.



MSMEs



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